Stanford Agentic Reviewer — Review Report

Availability Is Not Enough: When Symbolic Tools Help—and Hurt—LLMs on Planning Tasks (AAAI-27 submission, iter 1)

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Summary

This paper presents a rigorous empirical study of when access to sound PDDL tools (planners, validators, simulators) helps or hurts LLMs on planning tasks. Using a three-arm design that separates tool availability from an explicit one-sentence steering directive, strict end-to-end grading against deterministic oracles, and confidence intervals with a signed-significance rule, the authors evaluate five single-tool tasks across five open-weight models and two reasoning modes. The main findings are that tool utility is strongly regime-dependent, with *solve* and *simulate* being essentially impossible without tools, validation tasks improving over weak baselines (including a "rescue" on domain validation), and plan validation revealing that mere tool availability can significantly reduce success due to under-invocation, which a single steering sentence repairs; overall, invocation propensity, not capability or tool accuracy, is the bottleneck.

Strengths

Technical novelty and innovation

- Clear, well-motivated three-arm design that cleanly disentangles tool availability from explicit steering and enables precise attribution of effects.
- Elegant decomposition of success into P(call) $\times$ P(correct | call), which pinpoints invocation propensity as the binding constraint and explains counterintuitive failures (availability can harm).
- Signed-significance decision rule that distinguishes helpful from harmful effects and avoids misinterpreting large negative shifts as "just effects."
- Token "cost-of-pass" metric provides a practical readout of utility vs. cost and yields an actionable rule-of-thumb for deployment.

Experimental rigor and validation

- Strict end-to-end oracle-based grading across five tasks with balanced positives/negatives where appropriate and a large number of trials per cell.
- Confidence intervals for all reported proportions, plus MOVER intervals for cross-mode differences; explicit accounting for paraphrase clustering and design effect.

- Anonymized-domain contamination control that is carefully analyzed and convincingly null under the clean (think=off) condition.
- Robustness analysis that exposes and mitigates shared decode-budget confounds in reasoning mode via budget-insensitive summaries.

Clarity of presentation

- Well-structured exposition that foregrounds the central question, methodology, and the rationale for design choices; strong figures and failure-mode breakdowns aid interpretability.
- Transparent limitations section that anticipates key objections (decode budget sharing, quantization, prompt differences) and positions claims accordingly.

Significance of contributions

- Delivers concrete, generalizable insights for tool-augmented LLMs: availability is not enough; explicit steering often determines success; tools pay off most where unaided ability is floored.
- Offers a more rigorous and cleanly controlled evaluation than prior work (e.g., Planning Copilot), establishing a higher standard for tool-use assessments in planning.

Weaknesses

Technical limitations or concerns

- Reasoning-mode (think=on) results are confounded by shared decode budgets; no cap-raised rerun is presented to remove this artifact.
- Largest models are INT4-quantized while smaller ones are 16-bit, introducing a potential confound in cross-model comparisons (especially for tool-calling propensity).
- The availability-vs-no-tools prompts differ by a small policy/format clause beyond tool presence, leaving a minor residual confound in the availability gap.

Experimental gaps or methodological issues

- Deterministic decoding (temperature 0) limits understanding of robustness to sampling variance; no sensitivity to alternative sampling policies or minor prompt variants beyond the single steering sentence.
- Statistics acknowledge paraphrase clustering but rely primarily on Wilson intervals with a post hoc inflation check; more principled clustered or mixed-effects inference would strengthen claims.
- The evaluation is restricted to single-tool tasks; selection among multiple relevant tools and tool-chaining—more representative of agentic settings—are untested.

Clarity or presentation issues

- While well-written overall, some metric choices (e.g., using pooled accuracy for significance on validate_domain but discussing balanced accuracy in the narrative) may feel uneven; a unified statistical treatment would be cleaner.
- Steer-sentence content is not deeply varied or ablated; readers may want to know how sensitive the propensity effect is to specific wording and to distractor instructions.

Missing related work or comparisons

- Empirical scope excludes direct head-to-head with recent formalizer pipelines and multi-agent orchestration frameworks that also integrate symbolic tools, limiting external comparability.
- Frontier models and closed-weight systems are out of scope; while understandable, it leaves open whether the observed propensity dynamics persist at the frontier.

Detailed Comments

Technical soundness evaluation

- The three-arm design is appropriate and thoughtfully executed; isolating availability from steering is the right abstraction for single-call tool use.
- End-to-end oracle grading eliminates LLM-as-judge biases and ensures correctness is measured where it matters. The no-partial-credit policy for trajectories is stringent but justified and clearly reported with failure-type decomposition.
- The signed-significance rule is a good guardrail that aligns statistical decisions with the research question (help vs. harm), avoiding common misreads of negative but significant effects.
- The invocation decomposition is insightful and necessary: it shows nearly all variance arises from P(call), not P(correct | call), which is consistently high when tools are used.

Experimental evaluation assessment

- Large per-cell trial counts and balanced instances (where applicable) are strong; inclusion of paraphrase variants is a nice touch that probes linguistic robustness.
- The contamination control via symbol anonymization is a valuable addition and convincingly null under think=off, supporting the central claims.
- Reasoning mode analysis is candid: the shared budget induces baseline collapses. The use of budget-insensitive summaries (realizable benefit and robust floor) is careful, but a cap-raised rerun would remove lingering doubt.
- The token "cost-of-pass" analysis is practical and well-formulated; reporting both tokens/trial and success ratios clarifies where the tool is economical vs. a luxury.

Comparison with related work (using the summaries provided)

- Relative to Benyamin et al. (Planning Copilot), this paper substantially tightens evaluation: end-to-end oracle scoring, confidence intervals, steering-vs-availability separation, contamination control, and cost analysis. This directly addresses limitations noted in that work.
- The findings resonate with the LLM-Modulo thesis (Kambhampati et al.), showing tools deliver reliability only when orchestration actively routes work to them; merely exposing tools is insufficient.
- Compared to broader planning benchmarks (e.g., Valmeekam et al.), this study does not position LLMs as planners but as tool users, which explains why P(correct | call) is high and complements prior evidence that pure LLM planners lag classical planners. The results align with other works advocating hybridization and external validation.
- Recent formalizer and domain-induction efforts are complementary; this paper's focus is not on generating PDDL but on using existing formal tools with tight evaluation. A future head-to-head with formalizer pipelines under identical end-to-end criteria would be valuable.

Discussion of broader impact and significance

- The work advances evaluation methodology for LLM tool use and provides actionable deployment guidance: add explicit steering when exposing sound tools, and expect the biggest returns in tasks beyond unaided capability.
- The identification of invocation propensity as a volatile, prompt- and model-dependent behavior has implications for tool-use training, prompting, and alignment—suggesting that calibration or policy shaping may be as crucial as raw model capability.
- The token-economics analysis is important for practitioners weighing tool integration costs, underscoring that "tools pay where the baseline is floored."

Questions for Authors

1. How sensitive are the propensity effects to the exact wording of the single steering sentence? Did you test multiple semantically similar or dissimilar phrasings, including potential distractors, and do results persist?
2. For the availability-vs-no-tools contrast, can you provide an ablation where the policy/format clause is held constant to fully isolate tool presence, or otherwise estimate its contribution?
3. Could you run a targeted, higher-cap decode-budget experiment for think=on on a subset of tasks to confirm that the robust floor fully captures the true cross-mode picture?
4. How stable is P(correct | call) across domains and longer/rarer failure cases (e.g., numeric planning with more complex effects)? Any evidence that some domain classes erode accuracy-when-called?
5. Would clustered or mixed-effects logistic models (with instance and paraphrase random effects) materially change your significance conclusions relative to Wilson/MOVER intervals?
6. Did you attempt a "multiple relevant tools present" setup (e.g., validator and simulator both available) to explicitly measure tool selection under ambiguity?
7. Beyond anonymization, did you consider obfuscating structural features (e.g., permuting action arities or introducing type indirections) to probe deeper for contamination or surface-regularity dependence?
8. How portable are your findings to agentic multi-call settings (e.g., iterative refinement, replanning), where invocation choices are repeated and compounded over horizons?
9. Can you comment on whether quantization influenced tool-calling behavior (e.g., via logit-lens diagnostics or a small 16-bit run of the 35B on key cells)?
10. Will you release the full harness, fixtures, anonymization scripts, and prompts to enable exact replication, including the MCP tool specifications and schema?

Overall Assessment

This is a strong and timely empirical paper that sets a high bar for evaluating LLM tool use in planning. The study is carefully designed, transparently analyzed, and yields crisp, actionable insights: tools are indispensable on tasks beyond unaided reach; explicit steering is often necessary; and invocation propensity is the core bottleneck. While some methodological limitations remain (decode-budget confounds in think=on, minor prompt differences across arms, quantization confounds, and a focus on single-tool settings), they are acknowledged and largely mitigated, and they do not detract from the central findings. The work advances methodology relative to prior systems papers and clarifies an important deployment lesson for LLM-Modulo frameworks. I view this as a valuable contribution suitable for AAAI.